Worksheet 1-15 A
Limits at Infinity and Horizontal Asymptotes #1
AP Calculus Name_____

Directions: Find the following limits, if possible.

1. $\lim\limits_{x\to\infty} \dfrac{x^2-2x+4}{3x^2-1}$	2. $\lim\limits_{x\to\infty} \dfrac{4x-5x^3+1}{2x^3-5x^2+7}$	3. $\lim\limits_{x\to\infty} \dfrac{6x^2-1}{2x^3-4x+1}$
4. $\lim\limits_{x\to\infty} \dfrac{(3-4x)(2x^2+1)}{2x^3+4x^2-3}$	5. $\lim\limits_{x\to\infty} \dfrac{4x^2-2x+5}{8x+11}$	6. $\lim\limits_{x\to-\infty} \dfrac{x^2-16}{5x^2-2x}$
7. $\lim\limits_{x\to\infty} \dfrac{\sqrt{9x^2+4x-2}}{6x+4}$	8. $\lim\limits_{x\to\infty} \dfrac{5x-12}{\sqrt{3x^2-1}}$	9. $\lim\limits_{x\to-\infty} \dfrac{\sqrt{16x^2+2x-3}}{8x+4}$
10. $\lim\limits_{x\to\infty} \dfrac{x^2+3^x}{3x^2-1}$	11. $\lim\limits_{x\to\infty} \dfrac{e^x-1}{4-3e^x}$	12. $\lim\limits_{x\to-\infty} \dfrac{4-2^x}{2^x+3}$
13. $\lim\limits_{x\to\infty} \dfrac{8-4x-6x^2}{(x-3)(4-2x)}$	14. $\lim\limits_{x\to-\infty} \dfrac{(3+2x^2)(8+4x^3)}{3x^5+4x^3-2x}$	15. $\lim\limits_{x\to-\infty} \dfrac{4^x-7}{8+4^x}$

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16. $\lim_{x\to\infty} \frac{5x^3+3x+2}{7x^4+5}$

17. $\lim_{x\to-\infty} \frac{\sqrt{5x^2+6}}{8+4x}$

18. $\lim_{x\to\infty} \frac{3(2^x)+4}{5-2^x}$

19. $\lim_{x\to\infty} \frac{x^{100}}{3^x}$

20. $\lim_{x\to\infty} \frac{2x^2+4}{\sqrt{25x^4+x+3}}$

21. $\lim_{x\to-\infty} \frac{e^x+3}{5-2e^x}$

Directions: Find any horizontal asymptotes for the following functions.

22. $f(x) = \frac{x^2-2x+5}{3x^2+1}$

23. $g(x) = \frac{3x^2+4x}{x^3-4x^2-2}$

24. $h(x) = \frac{\sqrt{4x^2+5}}{3x-1}$

25. $k(x) = \frac{3^x-4}{2+3^x}$